Theme Detection - updates

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Today we want your input on the design trade-offs.

Goal: surface accelerating investment themes by combining news and ETF signals.

Topic detection: classical methods (1/2)

Bag of Words. Documents → vocabulary → document–term matrix $\mathbf{X}$.

Example docs: `cats chase mice` | `dogs chase cats` | `stocks and markets rise`

Hard assignment (K-means, hierarchical clustering): one label per doc
(D1, D2 → Animals; D3 → Finance). *Limitation:* real text is multi-topic.

Matrix factorization: $\mathbf{X} \approx \mathbf{WH}$ ($\mathbf{W}$ doc–topic, $\mathbf{H}$ topic–word) ⇒ soft mixtures, latent structure. Methods: LSA (SVD), NMF.

Document–term matrix

Doc	cats	chase	mice	stocks
D1	1	1	1	0
D2	1	1	0	0
D3	0	0	0	1

pLSA / LDA (probabilistic mixtures):

$$P(w \mid d) = \sum_t P(w \mid t)\, P(t \mid d)$$

LDA adds Bayesian priors: each document has a distribution over topics; each topic over words. Became the classical default (interpretable, scalable).

Topic detection: embeddings and BERTopic (2/2)

Shift: word counts $\rightarrow$ semantic vectors (Word2Vec, BERT, Sentence-BERT). Each document $d_i \in \mathbb{R}^{768}$; similar meaning $\Rightarrow$ nearby in space.

BERTopic pipeline: documents $\rightarrow$ embeddings $\rightarrow$ UMAP $\rightarrow$ HDBSCAN $\rightarrow$ c-TF-IDF keywords. Combines transformer embeddings, nonlinear dimensionality reduction, density clustering, and class-based TF-IDF for interpretable labels.

No fixed K: topics are *dense regions* in embedding space—HDBSCAN finds stable clusters of semantically similar documents rather than prescribing the number of topics upfront.

Classical

Topics are **word frequency distributions** inferred from $\mathbf{X}$:
hard labels, factorized latent factors, or generative models (pLSA, LDA).

Modern (BERTopic)

Topics are **regions in semantic embedding space**: geometry of meaning drives clustering; keywords summarize each region post hoc.

Open questions: what is a theme?

Before we fix the pipeline, we need shared answers (or explicit trade-offs) on:

1. What is a **theme**?
2. Is a theme the same as a **topic** (in the NLP sense)?
3. Is a theme a **cluster of news**—or an **abstraction** on top of a cluster?
4. Does it have a clear **geographical** location or scope?
5. Does it have a clear **temporal** placement or scope?
6. What **level of granularity** should it have?
7. For each theme we surface: can it be **monetized** (mapped to investable exposure)?